

SOUMYA SAHNI

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Recent Information Technology graduate with hands-on experience in Data Science, Machine Learning, Artificial Intelligence, and Data Analytics. Skilled in Python, SQL, Power BI, and machine learning frameworks for data analysis, predictive modelling, visualization, and intelligent system development. Experienced in exploratory data analysis, NLP, forecasting systems, and dashboard development using real-world datasets. Passionate about leveraging AI and data-driven insights to solve business and analytical problems.

Education

Manipal University

August 2025 – Present

Master of Computer Applications in AI and Data Science (online)

Narsee Monjee Institute of Management Studies (NMIMS)

July 2025 – Present

Master of Business Administration in Operations and Data Science Management (online)

Amity University

August 2022 - July 2025

Bachelor of Sciences in Information Technology

Uttar Pradesh, India

Ahlcon Public School

April 2008 - May 2022

PCMB

Delhi, India

Internship Experience

Unified Mentor

May 2024 – July 2024

Data Analyst Intern

Gurugram, Haryana

- Analyzed business datasets to identify sales trends and customer insights.
- Built Power BI dashboards to support reporting and decision-making.
- Performed data cleaning and EDA using Python libraries.

Personifywy

July 2023 – August 2023

Data Science Intern

Bengaluru, Karnataka

- Contributed to AI-powered chatbot development using Python and NLP.
- Built text preprocessing and feature extraction pipelines for better response accuracy.
- Assisted in optimizing conversational flow and personalized interactions.

Projects

End-to-End Retail Analytics & Demand Forecasting System

Python | SQL | Machine Learning | Power BI | FastAPI

- Performed exploratory data analysis on retail sales datasets to identify revenue trends and purchasing patterns.
- Built machine learning models for product demand forecasting and future sales prediction.
- Developed interactive Power BI dashboards and deployed prediction APIs using FastAPI for real-time forecasting insights.

Network Intrusion Detection System (NIDS)

Machine Learning | Python | Power BI

- Built a machine learning-based intrusion detection system using CICIDS2017 and KDD'99 datasets to identify cyber threats such as DDoS and brute-force attacks.
- Applied PCA for feature reduction and SMOTE for class balancing to improve model performance.
- Implemented ensemble ML models achieving 90–95% detection accuracy.
- Developed Power BI dashboards to visualize network anomalies and model performance metrics.

Sahni Realtors – Real Estate Analytics Platform

React.js | Node.js | MongoDB | Power BI | Data Analytics

- Developed a real estate platform integrating property listings with area growth and investment insight dashboards.
- Built analytics features showcasing location development trends, investment potential, and market insights for buyers.
- Integrated company performance metrics and historical business data to highlight Sahni Realtors' market credibility and client trust.

Customer Segmentation Using Machine Learning

Python | Scikit-learn | Power BI

- Applied K-Means clustering to segment customers based on purchasing behaviour and spending patterns.
- Analyzed customer data to identify high-value customer groups and revenue-driving segments.
- Built Power BI dashboards to visualize customer insights and support targeted marketing strategies.

LLM Document Question Answering System

Python | LangChain | FastAPI | Vector Databases

- Performed exploratory data analysis on retail sales data using Python (Pandas, NumPy) to identify trends and patterns.
- Developed a Generative AI application enabling users to extract answers from PDF documents using LLMs.
- Implemented semantic search and Retrieval-Augmented Generation (RAG) pipelines using vector databases and LangChain.
- Built FastAPI-based REST APIs for real-time AI-powered document querying and response generation.

Technical Skills

Programming Languages

Python, SQL, Java, JavaScript

Data Science & ML: Machine Learning, NLP, EDA, Predictive Modeling, Scikit-learn, Pandas, NumPy

Visualization: Power BI, Matplotlib, Seaborn

Technologies & Tools

Frameworks & Tools: FastAPI, Flask, LangChain, Git, GitHub, Jupyter Notebook

Databases

MySQL, PostgreSQL, MongoDB